LAB 5: STATIC CODE ANALYSIS

Event Manager — Issue Identification and Resolution

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Step 3: Identify and Document Issues

Issue	Type	Line	Description	Fix Approach
Mutable default argument	Bug	9	The default value guests=[] could be shared across function calls.	Changed the default to None and initialized a new list inside the function.
Unsafe pickle deserialization	Security	2, 51	pickle.load() can execute malicious code when loading untrusted data.	Replaced pickle-based persistence with JSON serialization and deserialization.
Shell command execution	Security	56	os.system() constructed a shell command using a variable event name, creating an injection risk.	Removed the unused shell-based printing helper.
Pseudorandom selection	Security	72	random.choice() triggered a Bandit warning about using a standard pseudorandom generator.	Replaced random.choice() with secrets.choice().
Missing module/function docstrings	Code quality	Multiple	The module and functions did not have descriptive docstrings.	Added module-level and function-level docstrings.
Naming convention violations	Code quality	Multiple	Function and variable names used camelCase rather than snake_case.	Renamed functions and variables using snake_case.
Unused sorting argument	Code quality	39	sortOrder was declared but not used by the guest report function.	Implemented sorting based on the requested order.
Unnecessary else after return	Code quality	33	An else branch followed a branch that already returned.	Removed the unnecessary else and returned directly.
Whitespace violations	Formatting	61, 66, 75	Missing or incorrect whitespace violated Python style conventions.	Corrected spacing around commas, operators, and inline comments.
Insufficient blank lines	Formatting	79	The code did not have the required blank-line separation.	Added the required blank lines between definitions.

Verification Results

Tool	Final result
Pylint	10.00/10
Bandit	No issues identified
Flake8	No formatting violations reported

Commit History

1. Fix security issues identified by Bandit
2. Fix code quality issues identified by Pylint
3. Fix formatting issues identified by Flake8

The final code was checked using all three static analysis tools.